

MABIOR DENG KUER MABIOR

PhD Student, Aviation and Aerospace Technology
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RESEARCH PROFILE

PhD researcher specializing in Advanced Air Mobility (AAM), airport sustainability, aviation safety systems, and transportation resilience. Current research evaluates the safety and sustainability implications of integrating AAM operations into large U.S. hub airports using systems-based and GIS-enabled decision-support frameworks. Research interests include airport infrastructure planning, transportation systems analysis, sustainability assessment, aviation policy, and airspace integration.

EDUCATION

Purdue University, West Lafayette, Indiana, USA

Doctor of Philosophy (PhD), Aviation and Aerospace Technology | 2024–Present

Track: Computational Science and Engineering (CSE)

GPA: 3.94/4.00

Dissertation: Sustainability assessment of Advanced Air Mobility integration at large U.S. hub airports using the EONS framework and GIS-enabled decision-support methods.

Honors: Frederick N. Andrews Fellowship

University of South Wales, Treforest, United Kingdom

Master of Science (MSc), Aviation Engineering and Management | 2020

Graduated with Distinction

Dissertation: Evaluating the Implementation of Total Productive Maintenance (TPM) within the East African Aviation Maintenance, Repair, and Overhaul (MRO) Industry

Honors: Chevening Scholarship; Faculty Certificate of Excellence

Shenyang Aerospace University, Shenyang, China

Bachelor of Engineering (BE), Aeronautical Engineering | 2016

Dissertation: Finite Element Simulation of Skin Stretch Forming

Honors: Multiple Semester Academic Excellence Awards

PUBLICATIONS AND RESEARCH OUTPUTS

Manuscripts Under Review

Mabior, M. D. K., Marete, C.K., & Hupy, J. P. (under review). *Advanced Air Mobility at Large Hub Airports: A Critical Investigation of Analytical Fragmentation, Metropolitan Readiness, and Policy Constraints. Transportation Research Part A.*

Yu, J., Mabior, M. D. K., Jin, S., & Mary E. Johnson (under review). *Cognitive and Human-Automation Challenges in UAS-AAM Integration: A Thematic Review. Journal of Aviation/ Aerospace Education & Research.*

Peer-Reviewed Conference Publications

Mabior, M. D. K., & Caroline K. Marete (2025, June). *Aviation knowledge and sustainability: Epistemological gaps in decarbonization across the Global North and South.* Paper presented at the 2025 ASEE Annual Conference & Exposition, Montreal, Quebec, Canada. <https://doi.org/10.18260/1-2--55294>

Conference Presentations

Mabior, M. D. K., & Caroline K. Marete (2025, March). Navigating the Perils: Operational Safety in Conflict Zones. Presentation delivered at the 2025 Purdue Road School Conference & Expo, West Lafayette, Indiana, USA.

Technical and Institutional Reports

Mabior, M. D. K., Safi, M., & Kochovski, A. (2021). Information paper on ICAO's High-Level Conference on COVID-19 (HLCC 2021) Annex 19 related outcomes: Presented at the Safety Management Panel (SMP/5). International Civil Aviation Organization, Montreal, Canada.

RESEARCH EXPERIENCE

Graduate Researcher

Purdue University, West Lafayette, Indiana, USA | 2024–Present

Advisor: Dr. Caroline K. Marete

- Conduct research on Advanced Air Mobility (AAM) integration at large U.S. hub airports using systems-based and GIS-enabled decision-support frameworks.
- Develop interdisciplinary research integrating airport operations, transportation systems analysis, aviation policy, and geospatial decision-support methodologies.
- Contribute to peer-reviewed journal manuscripts, conference publications, and scholarly presentations related to aviation and aerospace systems and transportation sustainability.
- Perform systematic literature reviews and thematic analyses on aviation safety, operational sustainability, and emerging transportation research topics.

Research Associate

International Civil Aviation Organization, Montreal, Canada | 2021

- Contributed to the development of a technical information paper addressing Annex 19 safety management outcomes from the ICAO High-Level Conference on COVID-19 (HLCC 2021).
- Supported aviation safety and Safety Management System (SMS) discussions presented to the ICAO Safety Management Panel (SMP/5).
- Participated in collaborative analysis of operational aviation safety considerations associated with global pandemic recovery planning and international air transport operations.

TEACHING EXPERIENCE

Teaching Assistant

University of South Wales, Treforest, United Kingdom | 2019–2020

- Assisted with teaching and instructional support activities in aviation engineering and related technical coursework.
- Facilitated laboratory sessions, coursework discussions, and student engagement activities in engineering and aviation-related subjects.
- Provided academic support and technical guidance to students on engineering problem-solving and coursework concepts.
- Contributed to instructional activities designed to enhance student understanding of aviation engineering principles and applied technical concepts.

PROFESSIONAL EXPERIENCE

Head of Safety (AOC & AMO)

Mission Aviation Fellowship, South Sudan | 2023–2024

- Led aviation safety oversight activities associated with Air Operator Certificate (AOC) and Approved Maintenance Organization (AMO) operations.
- Supported implementation and continuous improvement of Safety Management System (SMS) processes, operational risk assessment activities, and aviation safety compliance initiatives.
- Coordinated operational safety monitoring, safety reporting, and risk mitigation activities in complex operational environments.
- Contributed to aviation safety planning, regulatory coordination, and operational safety assurance activities supporting humanitarian aviation operations.

Quality, Safety & Security Manager

Mission Aviation Fellowship, South Sudan | 2022–2024

- Supported organizational quality assurance, aviation safety, and operational security activities within humanitarian aviation operations.

- Assisted in implementation of safety policies, operational compliance procedures, and risk management initiatives.
- Participated in safety auditing, operational monitoring, and continuous improvement activities related to aviation operations and maintenance oversight.
- Contributed to safety awareness initiatives and operational coordination activities supporting aviation safety and organizational resilience.

Airworthiness Inspector

South Sudan Civil Aviation Authority, Juba, South Sudan | 2017

- Assisted in aviation airworthiness oversight and regulatory compliance activities related to aircraft operations and maintenance.
- Supported technical inspection, documentation review, and regulatory monitoring activities associated with civil aviation safety oversight.
- Participated in operational and maintenance compliance coordination activities supporting aviation safety and regulatory assurance.

HONORS, FELLOWSHIPS, AND AWARDS

Frederick N. Andrews Fellowship

Purdue University | 2024–Present

Chevening Scholarship

Foreign, Commonwealth & Development Office (FCDO), United Kingdom | 2019

Faculty Certificate of Excellence

University of South Wales | 2020

Multiple Academic Excellence Awards

Shenyang Aerospace University | 2012–2016

TECHNICAL SKILLS

GIS & Geospatial Analysis: ArcGIS Pro, spatial analysis, geospatial decision-support modeling

Statistical & Data Analysis: SAS, statistical modeling, regression analysis, probability, and statistical analysis

Simulation & Modeling: Arena Simulation Software, systems analysis, process modeling, transportation systems simulation

Research & Visualization Tools: VOSviewer, systematic literature review methodologies, bibliometric analysis, and scientific visualization

Programming & Computational Tools: Python, MATLAB, LATEX

PROFESSIONAL AFFILIATIONS

Royal Aeronautical Society

Member | London, United Kingdom

American Society for Engineering Education

Student Member | Washington, D.C., USA

The Royal Institute of International Affairs

Member | London, United Kingdom

Institution of Engineering and Technology

Member | London, United Kingdom

LANGUAGES

English: Fluent

Dinka: Native

Swahili: Native

Chinese (Mandarin): Conversational